



Dijkstra and Bellman–Ford

1 Dijkstra

Exercise 1. Give an example

```
1 for  $v \in V$  do
2    $d[v] \leftarrow \infty$ 
3    $\pi[v] \leftarrow \text{null}$ 
4 end
5  $d[s] \leftarrow 0$ 
6  $Q \leftarrow \text{Heap}(d[v_1], d[v_2], \dots, d[v_n])$ 
7 while  $Q$  is not empty do
8    $u \leftarrow \text{extract-min}(Q)$ 
9   for  $(u, v) \in E$  do
10    if  $d[v] > d[u] + w(u, v)$  then
11       $d[v] \leftarrow d[u] + w(u, v)$ 
12       $\text{Update}(Q, d[v])$ 
13       $\pi[v] \leftarrow u$ 
14    end
15  end
16 end
```

Algorithm 1: Dijkstra.

Detecting a Negative Cycle. Suppose that $v_1, v_1, \dots, v_k = v_1$ is a negative cycle. This means that:

$$\sum_{i=2}^k w(v_{i-1}, v_i) < 0$$

Assume that the algorithm doesn't detect the cycle, namely the condition at Line 14 is not satisfied for any of the edges. Then:

```

1 for  $v \in V$  do
2    $d[v] \leftarrow \infty$ 
3    $\pi[v] \leftarrow \text{null}$ 
4 end
5  $d[s] \leftarrow 0$ 
6 for  $i \in [n]$  do
7   for  $(u, v) \in E$  do
8     if  $d[v] > d[u] + w(u, v)$  then
9        $d[v] \leftarrow d[u] + w(u, v)$ 
10       $\pi[v] \leftarrow u$ 
11     end
12   end
13 end
14 for  $(u, v) \in E$  do
15   if  $d[v] > d[u] + w(u, v)$  then
16     return Negative Cycle
17   end
18 end

```

Algorithm 2: Bellman–Ford.

$$\begin{aligned}
& \forall v_i \ d[v] < d[u] + w(u, v) \\
& \Rightarrow \sum_{i=2}^k d[v_i] < \sum_{i=2}^k d[v_{i-1}] + \sum_{i=2}^k w(v_{i-1}, v_i) \\
& = \sum_{i=2}^k d[v_i] + \sum_{i=2}^k w(v_{i-1}, v_i) \\
& 0 \leq \sum_{i=2}^k w(v_{i-1}, v_i)
\end{aligned}$$